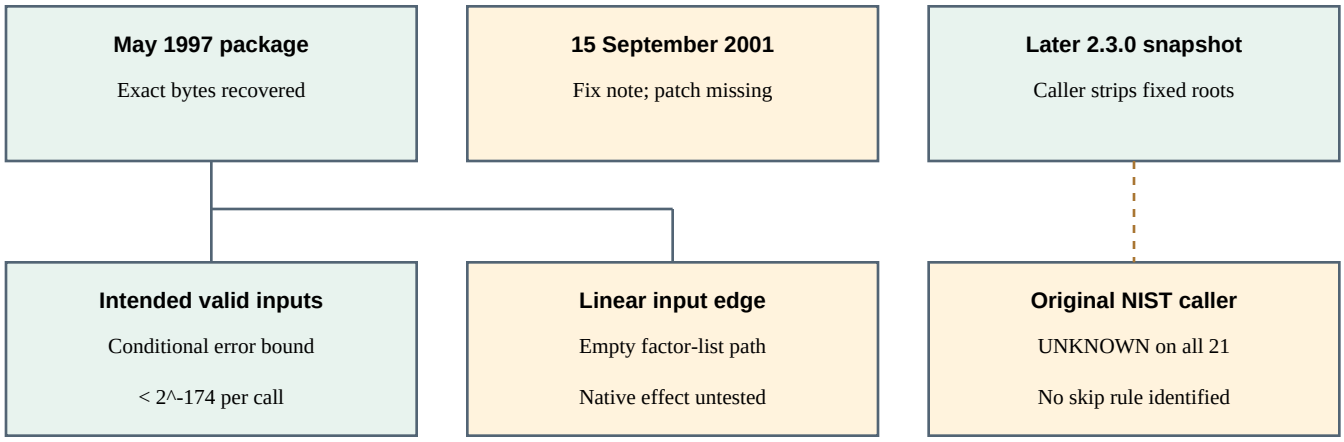
1997 degree routines: source evidence

28 October 2026 UTC | T-5518 | source-only audit; zero new native trials

Finding and limit. Exact source authenticates a later splitting-degree fix report and exposes an unguarded linear-input path. Neither identifies an original NIST call or explains any omission. All 21 historical applications remain undetermined.



Verified source facts and derived conditions; historical invocation remains missing.

Dated defect report. ChangeLog 15 September 2001 records an eco_gfp error fixed in compute_splitting_degree. Available Git history begins with a complete 2.3.0 import in 2013; it contains no exact 2001 patch. A separate 2019 power_x root-wrapper fix affects a filter absent from the 1997 split-root routine. Neither later defect is backdated.

Verified endpoints. The advertised May 1997 package is pinned by its 2,343,749-byte digest; README 1.3/header 1.2.0 discrepancy retained. Five original members and five later Git blobs are byte-verified. Four normalized helper/probabilistic arithmetic bodies match; tandem composition changes modulus initialization API, and the later wrapper accepts a supplied degree multiple. Textual comparison is not native equivalence.

Caller contract matters. Later SEA handles all-fixed/one-fixed cases separately, removes both linear factors in the Elkies two-fixed-root branch and then requests the common remaining degree. Its default is DONT_COMPUTE_SP_DEGREE; Atkin bound 50 applies when enabled. These are later source facts, not original NIST invocation evidence.

Sources: exact archived package capture 19970517195215; passagemath/upstream-LiDIA commit 23dac491f4cd96eb7de648a5cf90eb9ec77653c8, factoring.cc/eco_prime.cc/ChangeLog. Full URLs, original byte/Git hashes and read-line scopes are in SOURCE-MANIFEST.json, HISTORY.json and SOURCE-REVIEW.json.

Derived bounds and missing application

Conditional intended error bound. With squarefree equal-degree input, n>=2, p>n, correct arithmetic and independent uniform field draws, m=n/d trace components collide with probability at most m(m-1)/(2p). The separate min_poly projection adds floor(n/2)/p. For auxiliary primes ell<800 and n=ell+1<=798, the total is at most 79,800/p<2^-174 for these field sizes. Degree-one factors with n>=2 return directly when h=X. This is no measured frequency or historical RNG probability.

Static linear edge. For f=X-a the reduced Frobenius image is constant; the low-level h.is_x guard misses it. Integer degree 1 has an empty factor list, but recursion is entered with 0,-1, the splitter yields mid=-2 and a subsequent product includes index -1. This is an unguarded source path. No native crash or historical reachability is asserted; the later SEA caller handles the relevant cases earlier.

Complete cohort accounting. Full model/order/seed/ordinal/timestamp and exact UID are retained for every row in cohort-crosswalk.csv. The original degree-routine invocation and error policy are UNKNOWN on all 21. No failure is inferred from source absence.

Group	Exact identities	Historical application
Published	10	UNDETERMINED
Reconstructed omissions	11	UNDETERMINED
All six streams	21	UNDETERMINED

Custody checks. All 13 sealed metadata files, ten private source hashes including five Git objects, and every 21 identity binding pass. Relocated public metadata replay passes. These checks execute no native source or curve calculation. Raw commit 7ba50faa68 precedes report publication; unrelated files and frozen earlier records are preserved.

Next discriminator. Recover the pre-fix 2001 body or authenticated older caller with exact polynomial inputs and failure policy. A native linear-input control alone would establish only a low-level defect; original reachability remains separate. Any new curve trial needs a separately frozen protocol and all 21 controls/accounting. Existing signed-action/SEA graphs are unchanged; P-224 lead and full historical goal remain open.

Public replay: python3 verify_sources.py . Private bytes: append --sources. Complete symbolic derivation in PROBABILITY-BOUND.md; static path in LINEAR-EDGE.md. All source bodies stay ignored/private; metadata does not authenticate NIST use.